

Prediction in Policy Problems: *Foundations of AI-Based Decision Making*

Lecture 8 – AI4H

Andreas Bjerre-Nielsen

Where are we at?

- Part I: Foundations
 - *Tree-based, deep learning, transformers, LLMs*
- Part II: Policy and ML
 - *Fairness*
 - Prediction policy
 - AI for policy, other uses
- Part III: Causal ML
 - Causal trees and forests
 - Double ML

Overview of this lecture

- Fairness in practice
 - Aware vs blind models
 - Testing for improvability
- Prediction-guided policy
 - Policy vs prediction
 - Prediction policy cases
 - Medical diagnosis
 - Legal decisions by judges
 - College admission

Fairness in practice

Review of fairness (last week)

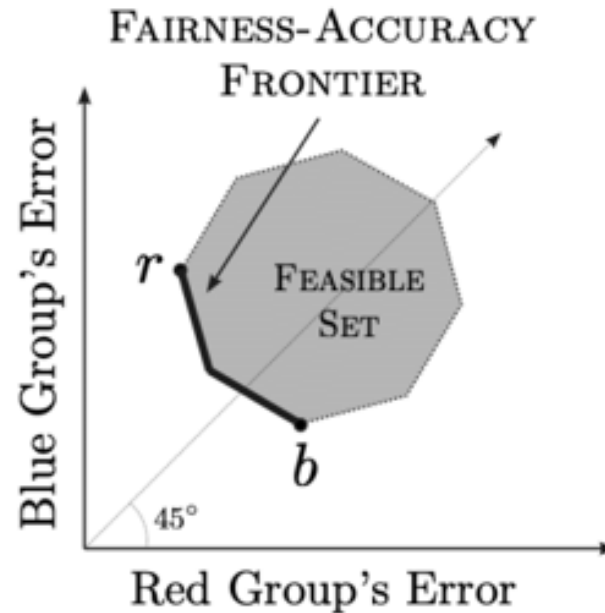
- Fairness measures:
 - equalized odds, demographic parity, calibration, etc.
- Introduced impossibility results
 - Fairness as calibrated prediction (statistical)
 - Fairness by equal odds (policy oriented)
- Now: moving from definitions → design and evaluation.

Liang et al. (2024, JPE)

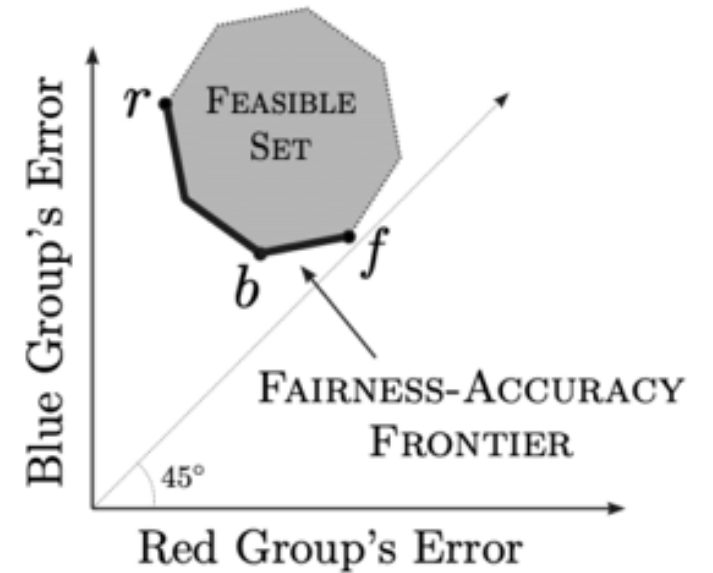
- Develops simple microeconomic framework for studying fairness
 - Two groups - blue and red
 - We care only about “errors”
 - Some weighted measure of false positive and negatives

Balancing fairness vs other consideration

- Is fairness a constraint?
- > Conclusion
 - group-balance: no
 - group-skewed: yes



(a) Inputs are Group-Balanced



(b) Inputs are Group-Skewed

FIGURE 1. The Fairness-Accuracy Frontier.

Special cases FA frontier

Group specific models

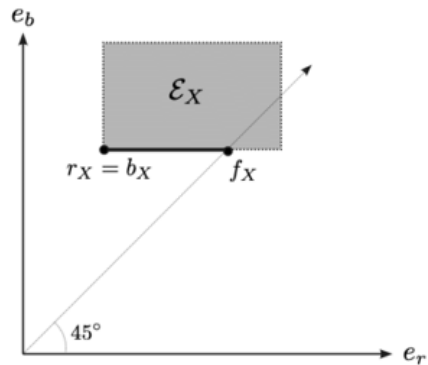
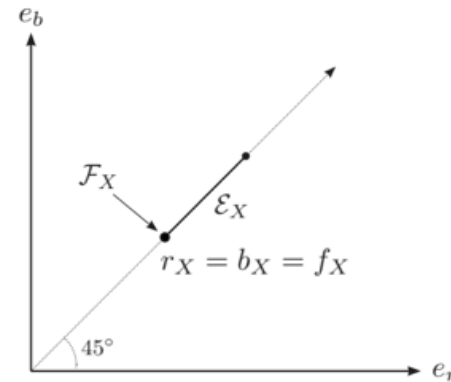
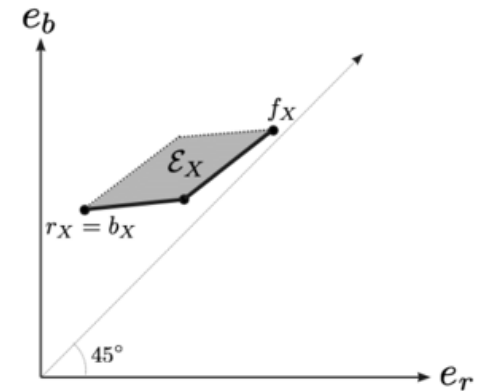


FIGURE 4. When group-specific algorithms are permitted, then the feasible set is a rectangle and the fairness-accuracy frontier is a line segment.

Independence criteria



(A) Strong Independence: $G \perp\!\!\!\perp (X, Y)$



(B) Conditional Independence: $G \perp\!\!\!\perp Y \mid X$

FIGURE 5. Special Cases

Input Design

- Definition of input design
 - How to process ethnic background or biological sex / gender
 - Objective minimize weighted errors for groups
 - Account for agents' actions in terms of revealing information
 - E.g., submitting information to banks for loan
- Examples of design policies
 - Banning an input
 - Coarsening bins
 - Add noise

Theoretical results

- Banning an input
 - Banning can be optimal for designer, but..
 - When considered feature, X , is group identity - never optimal to ban!
 - When considered feature is informative and not group identity, and group identity is included - never optimal to ban!

Empirical application

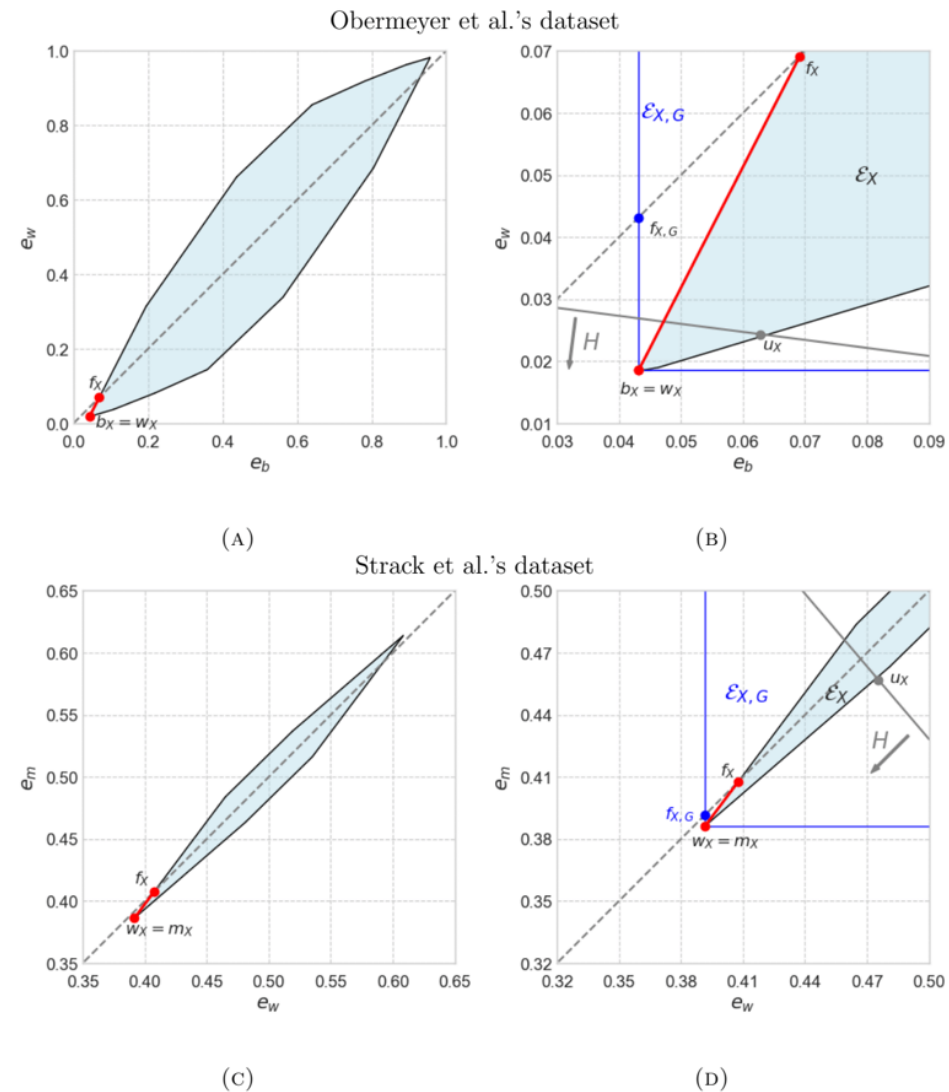


FIGURE 10. Estimated feasible set for the class of linear algorithms

Panels (A) and (C) display the estimated feasible sets for the datasets from Obermeyer et al. and Strack et al., respectively. In these panels, the black solid lines define the boundaries of the feasible sets $\mathcal{E}_X$, and the gray dotted lines serve as 45-degree reference lines. Panel (A) identifies b_X and w_X as the optimal points for Black and White patients, while Panel (C) marks w_X and m_X as the optimal points for women and men. Both panels highlight f_X as the fairness-optimal point. Panels (B) and (D) examine the augmented feasible sets $\mathcal{E}_{X,G}$ for the same datasets with covariate (X, G) , represented by blue solid lines. In these panels, $f_{X,G}$ indicates the fairness-optimal point for each dataset with (X, G) . The point u_X indicates the error pair chosen by the Utilitarian agent given no information, and the gray solid line in Panel (B) and (D) represents the Utilitarian's indifference curve through u_X .

Auerbach et al. (2025)

- Extend Liang et al. (2024)
- Key question:
 - Is the fairness/accuracy trade-off truly inevitable, or can we improve fairness and maintain accuracy?
- Consequence:
 - The “necessity defence” (that disparate impact is required for legitimate business interests) may be empirically testable.

Model setup

- The Null and Alternative Hypotheses
- They formally test whether an algorithm can either fairness and/or accuracy with harming the other!
 - H_0 : no algorithm improves either fairness or accuracy
 - H_1 : exist algorithm that improves
- Paper contribution – develop non-parametric statistic that enables inference (using cross-estimation)

Empirical application

- In their **healthcare algorithm application** (replicating Obermeyer et al., 2019):
 - They compute test stats for the status-quo model predicting health expenditures.
 - Bootstrap confidence intervals show rejection of H_0 : the algorithm is *improvable*.
 - They visualize results with a fairness-accuracy scatter plot (showing the status quo below the efficient frontier).

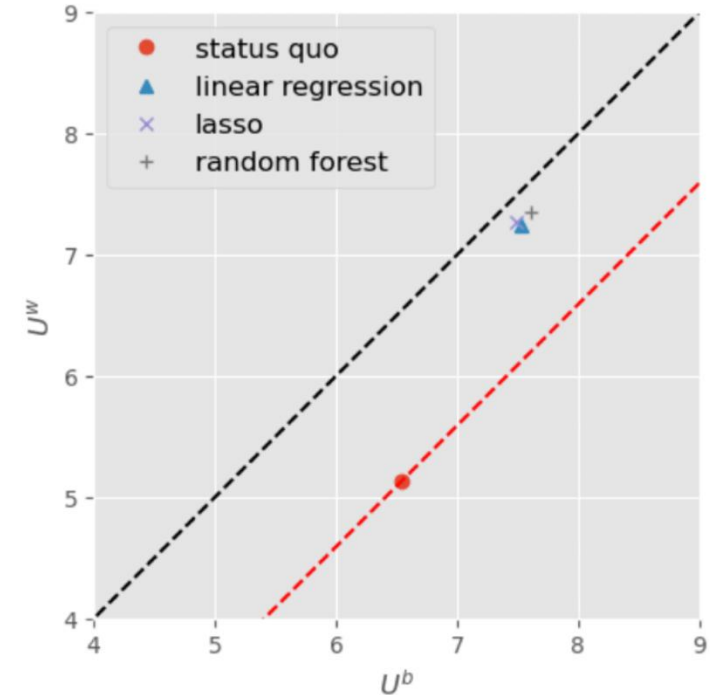


FIGURE 2. We report the average value of U^g (i.e., number of active chronic diseases conditional on automatic enrollment) for each group g across the $K = 7$ iterations of our procedure. The algorithm is more accurate for group g when U^g is larger; so, moving towards the upper-right quadrant corresponds to improvements in accuracy. The algorithm is more fair when its (U^b, U^w) pair is closer to the 45-degree line (corresponding to more balanced accuracy across the patient groups). This figure suggests that the candidate algorithms based on linear regression, LASSO, and random forest improve on both fairness and accuracy over the status quo algorithm used by the hospital.

Policy vs Prediction

Differences and overlaps in fundamental problems

Prediction vs Policy

- Predicting who will have a heart attack $\neq$ deciding who gets treatment
- AI systems predict outcomes, but decisions depend on goals and actions
 - Key question: *How can prediction be leveraged for decision?*
- Example domains: healthcare, finance, criminal justice, education

Prediction Problems

- Goal: Estimate relationships
 - Input/feature (X) $\rightarrow$ Outcome/target (Y)
 - How - determine function $f: X \rightarrow Y$
 - Objective: minimize prediction error (RMSE, log-loss, AUC)
 - Prediction. $\hat{y} = f(x)$

A simple policy problem

- Goal: Choose action that maximizes expected utility
 - Action / policy: a
 - (e.g., rain dance / who to hire)
 - Outcome: $y(a)$ [assumed affected by action, $A \rightarrow Y$]
 - (e.g., rain / productivity)
 - Utility/welfare: u [assumed affected by outcome, $Y \rightarrow U$]

A simple policy problem

- Optimal action: $a^* = \operatorname{argmax}_a(u(y(a)))$
 - Often we know what's good (utility function)
 - e.g., higher productivity / better standardized tests / improved health
 - Requires causal understanding how actions influence outcomes
 - > Need to estimate $\hat{y}(a)$

(generalized) Policy problems

- Goal: Choose policy π that maximizes expected utility or welfare for every given x !
- Policy $\pi(x)$: maps input/state, x , to an action/policy, a .
- Optimal policy: $\pi^*(x) = \operatorname{argmax}_{\pi} E[u(y(a))|x]$
 - Again, requires causal understanding how actions influence outcomes
 - estimate **heterogeneous treatment effect**, $Y(a|x)$

Kleinberg et al. (2015, AER)

- Assume now we have generalized functional form, $u(y, a)$
 - Action has two effects on utility, direct and through outcome y
- The dual effect (from total differentiation):

$$\frac{du(a, y)}{da} = \frac{\partial u(\cdot)}{\partial a} y + \frac{\partial u(\cdot)}{\partial y} \frac{\partial y}{\partial a}$$

- Direct effect $\frac{\partial u}{\partial a}$ depends on outcome y
 - A pure prediction problem!
 - Outcome effect $(\frac{\partial y}{\partial x})$, also affects utility indirectly $(\frac{\partial u(\cdot)}{\partial y})$
 - Again requires causal problem
- Example – bring rain coat or host garden party

Prediction policy cases

Medical diagnosis

Setup

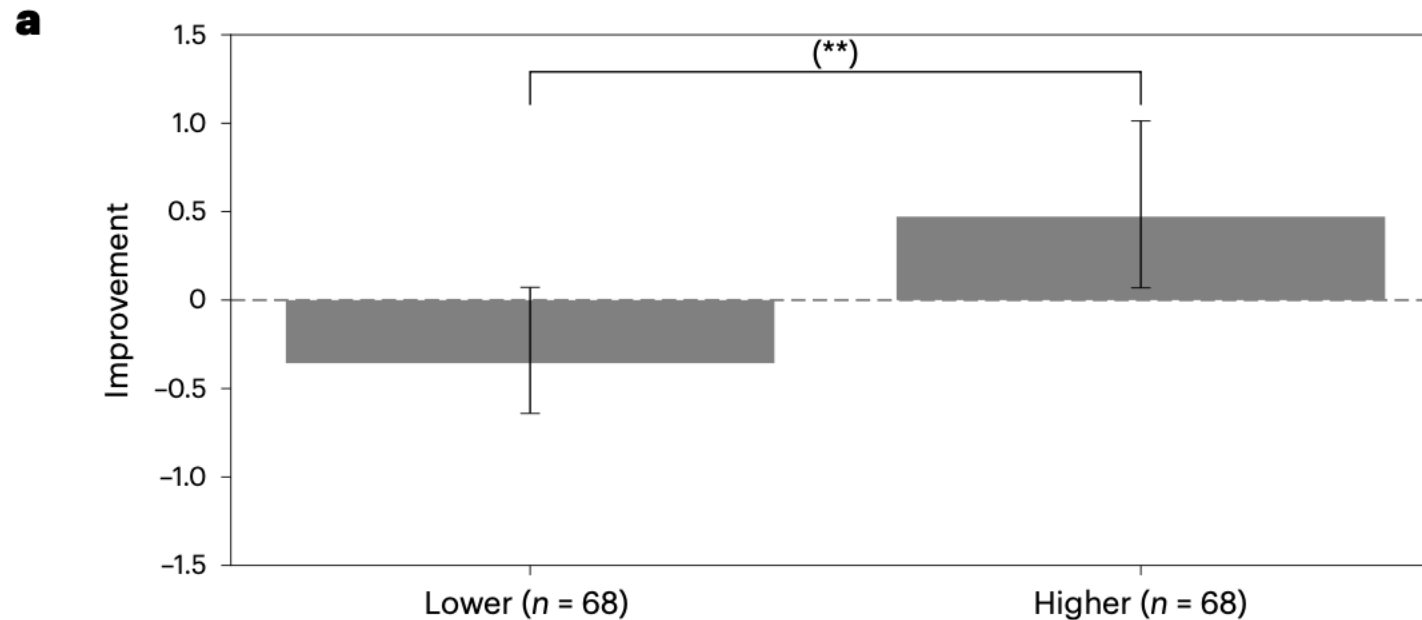
- We want to think about situations where we separate
 - medical diagnosis (outcome,)
 - medical treatment (action,)
- AI can be used for both
 - Focus here on *diagnosis* – prediction problem
 - Whether to give medical treatment – can use causal prediction methods
 - (part III of course)

Yu et al. (2024, Nat. Med.)

- Setup
 - 60 patient cases, radiologists examined
 - half without clinical histories
 - half with clinical histories
 - Among these random exposure to AI assistance
- Main outcome – correct diagnosis
 - measured as predicted diagnosis same as ground truth (yes vs. no)

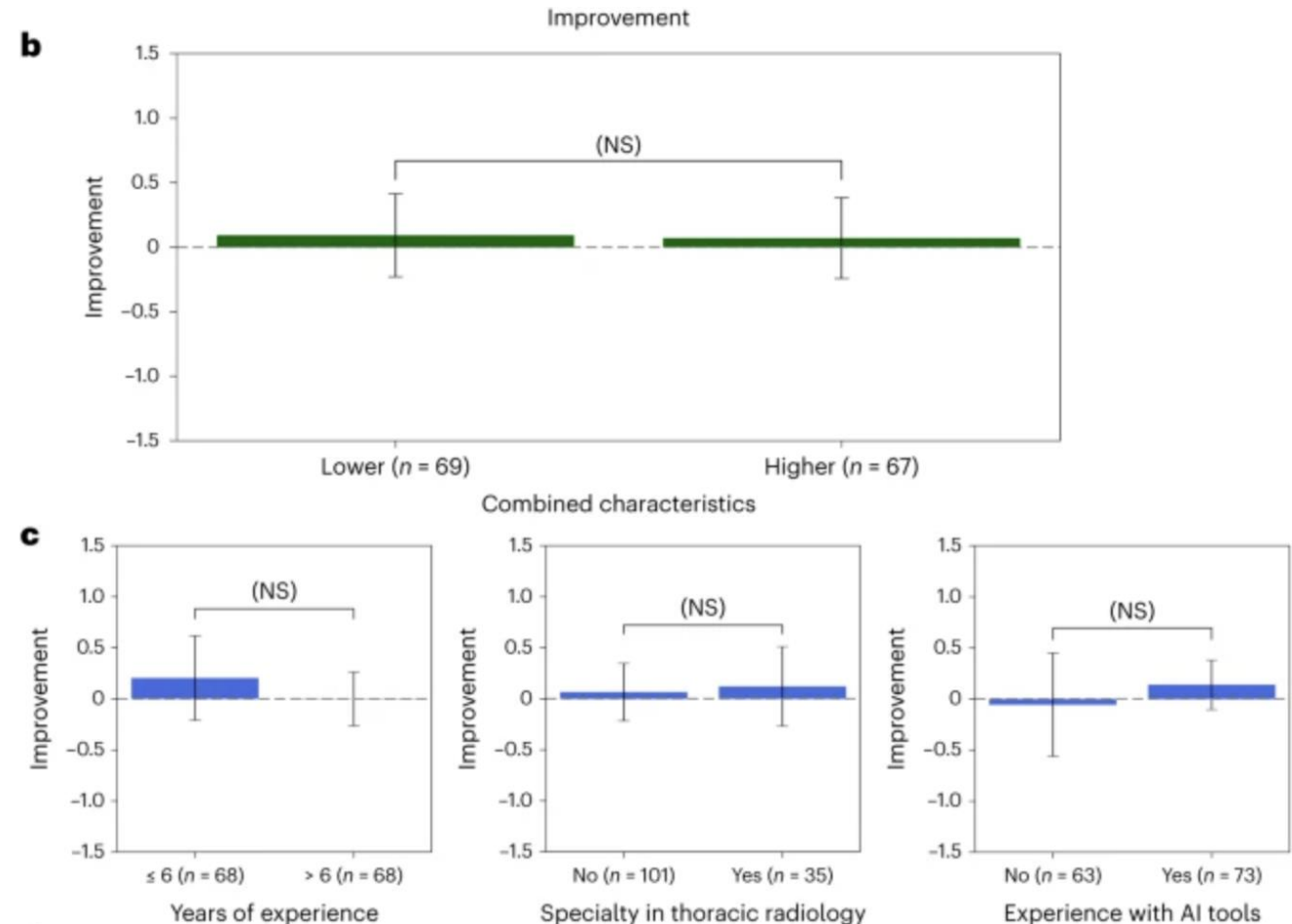
Heterogeneous impact

- Treatment effect of AI
 - Compute individual correctness with and without AI



Origin of heterogeneity - doctor characteristic

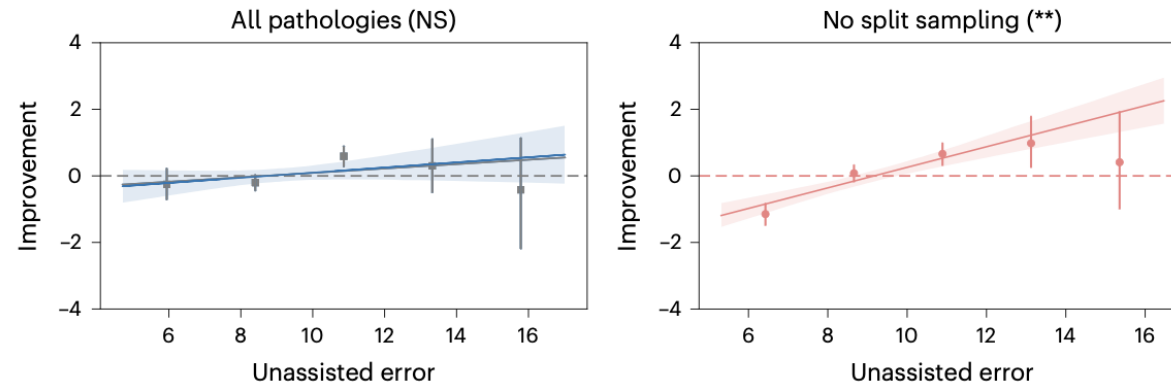
- Driven by doctor backgrounds?
 - years of experience, subspecialty and AI experience not predictive



Origin of heterogeneity - skill

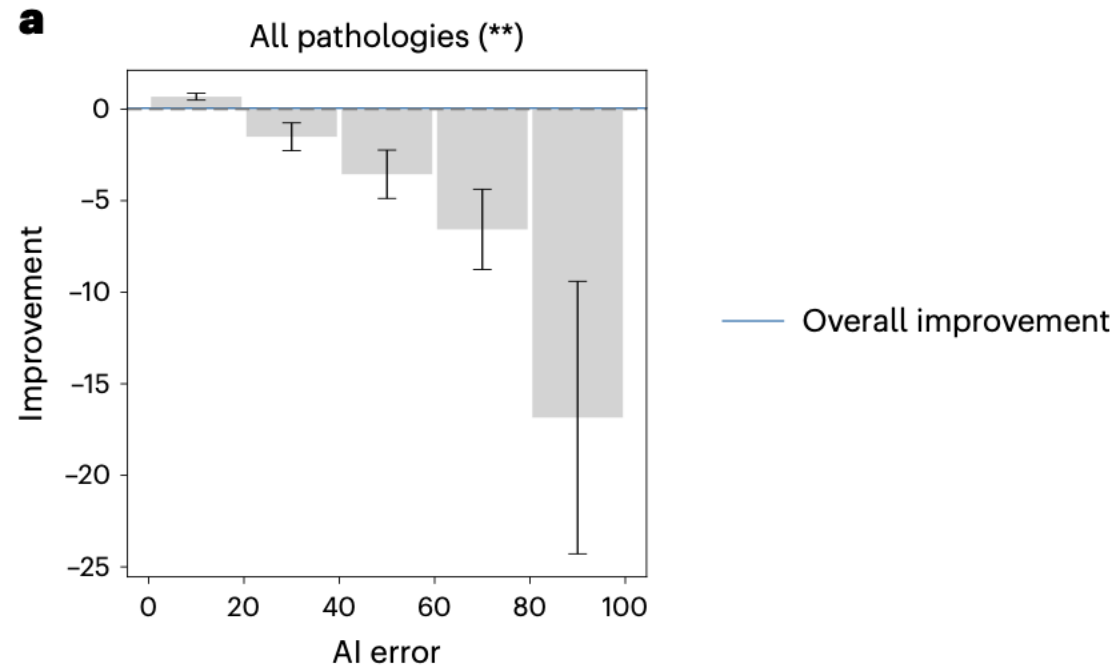
- Driven by doctor latent skill?
 - Measure as unassisted performance

a



Origin of heterogeneity – AI skill

- Driven by AI skill?



Summary

- Yu et al. find
 - Null effect of impact on AI on decision quality
 - Large heterogeneity in impact of AI!
 - Mainly driven by AI accuracy
- Next time – dive into Agarwal et al.

Medical diagnosis in the field

Predicting high-risk opioid prescriptions before they are given

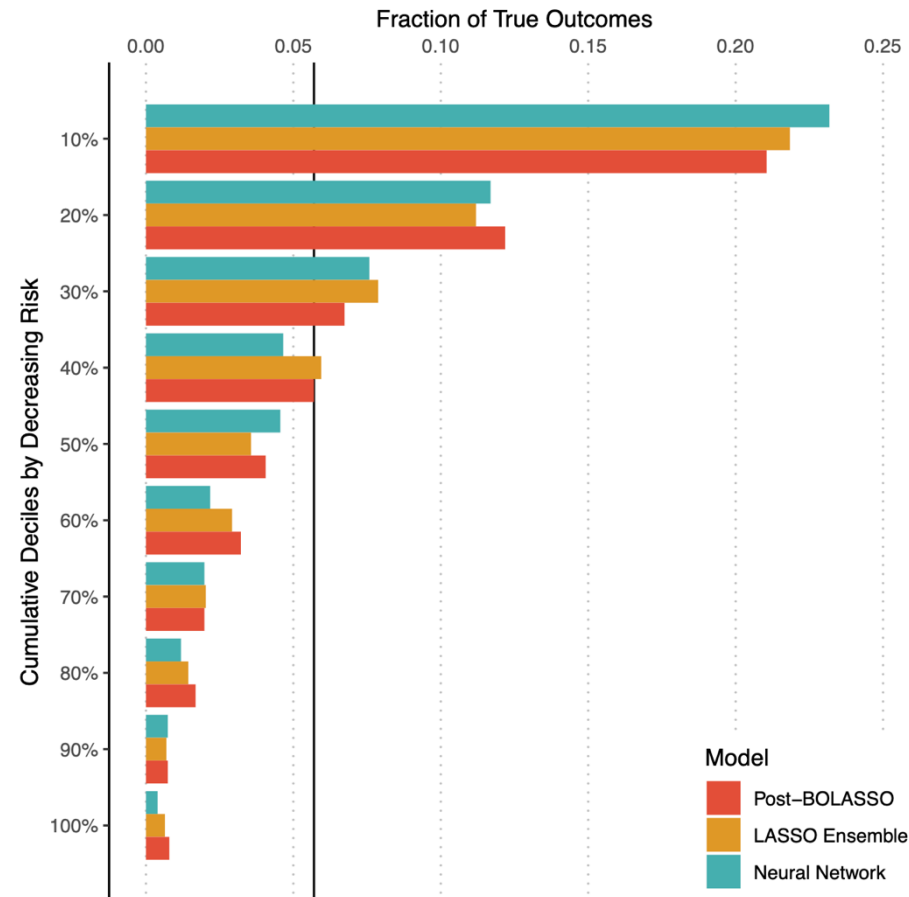
Justine S. Hastings , Mark Howison , and Sarah E. Inman [Authors Info & Affiliations](#)

Edited by Raj Chetty, Stanford University, Stanford, CA, and approved December 19, 2019 (received for review April 1, 2019)

January 14, 2020 | 117 (4) 1917-1923 | <https://doi.org/10.1073/pnas.1905355117>

Fig. S2. The fraction of true outcomes in the test sample. The vertical black line indicates the base rate of outcomes among the entire population, which is 0.057.

- Investigate opioid prescription by medical doctors
 - Could prediction flag pre-prescription risk?



Screening policy

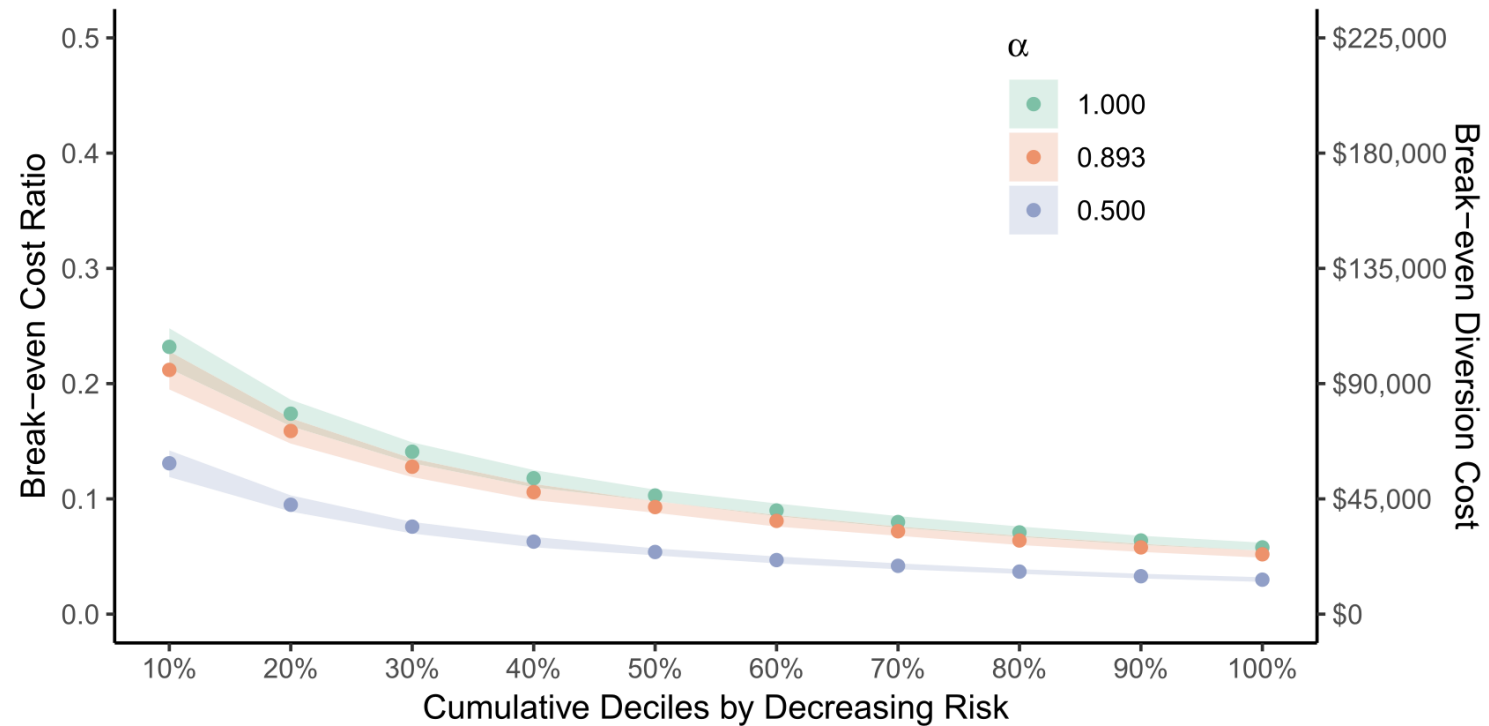


Fig. 2. The break-even cost ratio for three values of the efficacy rate α . The break-even cost ratio is the point at which the hypothetical policy becomes cost neutral. If the diversion cost is less than this ratio times the adverse outcome cost (estimated at \$450,000), then the policy will be net beneficial. Lower diversion costs are required to make the policy net beneficial among lower risk scores. Error bars indicate the 95% confidence interval calculated from 100 bootstrap replicates.

Summary

- Hastings et al. find
 - Prediction tools / AI can help screen for prescriptions
- Policy impact? Credibility
 - Could doctors be using unobserved information?
 - Could we replace doctor pre-scripture with algorithm?
- > Similar to legal decision

Legal decision as prediction policy?

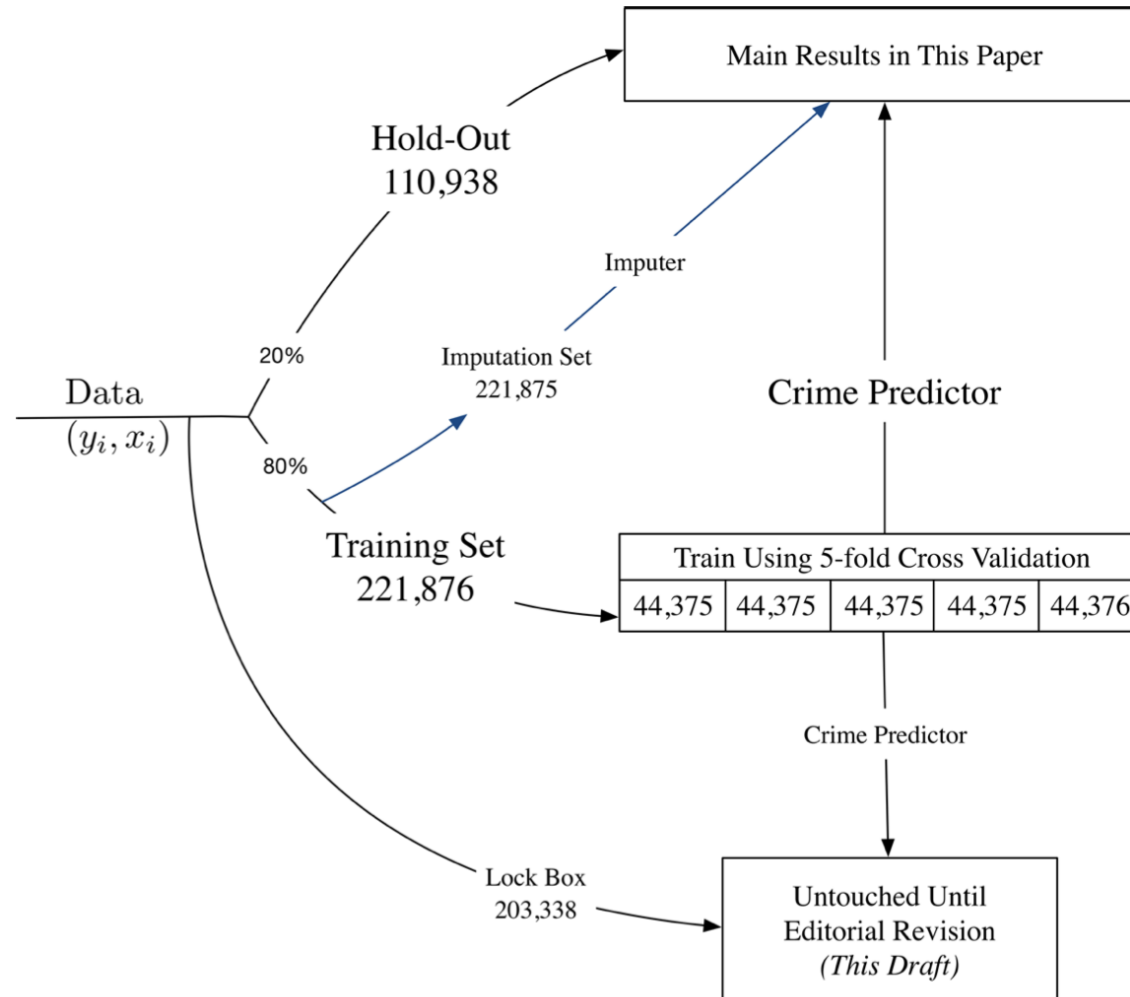
Prediction problem for judges

- Pretrial Bail Decisions
 - “Shortly after arrest, defendants appear at a bail hearing. In general, judges can decide to release the defendant outright (such as to release on recognizance set a dollar bail that must be posted to go free, or detain the defendant outright”
- Judges goals’ determine whether person will
 - fail to appear in court (FTA) or
 - be rearrested for a new crime
- Decision is independent of actual legal case!

Context and scope

- US nationally
 - 1 million people arrested annually!!
- New York City data 2008-13
 - cases # > 1 mill.
 - More than half with pre-trial decision

Training of legal algorithm



Training details

- Y: Outcome/target: crime
- X: Input – defendant characteristics
 - Demographics (only age)
 - Prior crime
 - Case material

Overall performance

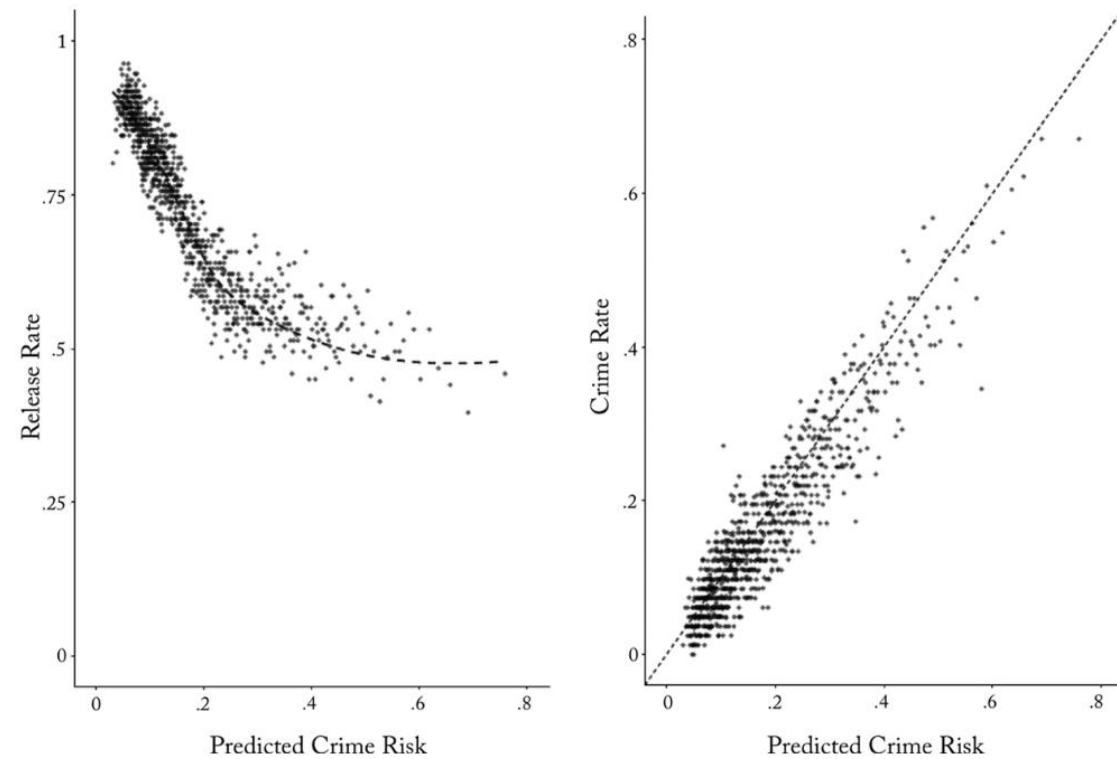


FIGURE II

How Machine Predictions of Crime Risk Relate to Judge Release Decisions and Actual Crime Rates

The figure shows the results of an algorithm built using 221,876 observations in our NYC training set, applied to the 110,938 observations in our test set (see Figure I). Both panels show the algorithm's predicted crime risk (defined here as predicted risk for failure to appear, or FTA) on the *x*-axis: each point represents one of 1,000 percentile bins. The left panel shows the release rate on the *y*-axis; the right panel shows the realized crime risk on the *y*-axis.

Selective labels problem

- Terminology
 - ρ_j - release rule of judge j
 - $m(x)$ - predicted risk from model
- We observe Y only if *released*!
 - $E[y|x, \rho_j = 0]$ cannot be measured
 - Could be that those retained by judges would have committed much more crime
 - Example: judges observe extra information, Z , e.g., gang tattoo

Policy evaluation – quasi-contraction

- “Contraction” approach
 - Release if and only if $\rho_j = 1$ and $m(x) < \kappa_j$,
- Evaluate outcomes as a function of $m(x)$

TABLE 11
COMPARING LOGISTIC REGRESSION TO MACHINE-LEARNING PREDICTIONS
OF CRIME RISK

Predicted risk percentile	ML/ logit overlap	Average observed crime rate for cases identified as high risk by:				
		Both ML and logit	ML only	Logit only	All ML cases	All logit cases
1%	30.6%	0.6080 (0.0309)	0.5440 (0.0209)	0.3996 (0.0206)	0.5636 (0.0173)	0.4633 (0.0174)
5%	59.9%	0.4826 (0.0101)	0.4090 (0.0121)	0.3040 (0.0114)	0.4531 (0.0078)	0.4111 (0.0077)
10%	65.9%	0.4134 (0.0067)	0.3466 (0.0090)	0.2532 (0.0082)	0.3907 (0.0054)	0.3589 (0.0053)
25%	72.9%	0.3271 (0.0038)	0.2445 (0.0058)	0.1608 (0.0049)	0.3048 (0.0032)	0.2821 (0.0031)

Notes. The table shows the results of fitting a machine-learning (ML) algorithm or a logistic regression to our training data set, to identify the highest-risk observations in our test set. Both machine-learning and logistic regression models are trained on the outcome of failure to appear (FTA). Each row presents statistics for the top part of the predicted risk distribution indicated in the first column: top 25% ($N = 20,423$); 10% (8,173); 5% (4,087); and 1% (818). The second column shows the share of cases in the top X% of the predicted risk distribution that overlap between the set identified by ML and the set identified by logistic regression. The subsequent columns report the average crime rate observed among the released defendants within the top X% of the predicted risk distribution as identified by both ML and logit, ML only, and logit only, and all top X% identified by ML (whether or not they are also identified by logistic regression) and top X% identified by logit.

Policy evaluation – leniency design

- Use random assignment locally (time and borough)
- Must check if risk is *not* associated with leniency within assignment cell

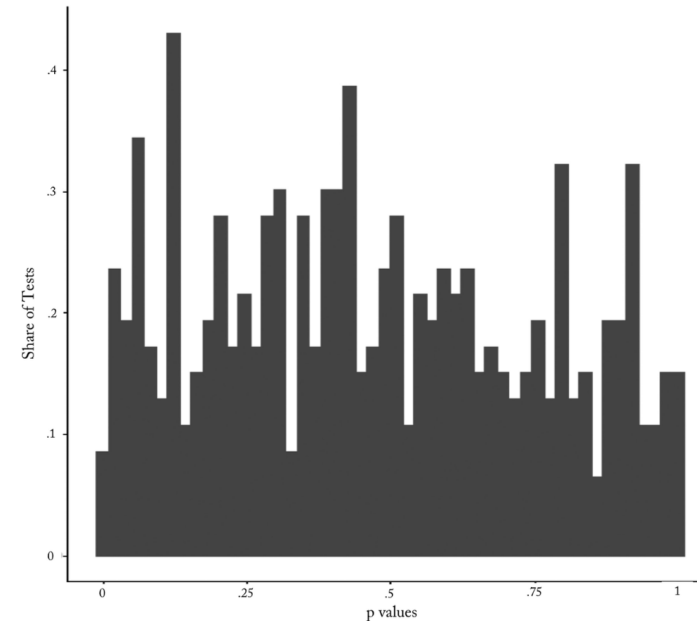


FIGURE III

Testing Quasi-Random Assignment of Defendants across Leniency Quintiles
Distribution of p -values for Balance Tests in Contraction Sample

The figure shows the distribution of p -values for balance checks in our contraction sample summarized in [Online Appendix Table A.3](#). We construct 577 borough, year, month and day of week cells in the New York City data where we have at least five judges. We then define judge leniency quintiles within each cell. We regress each defendant's predicted FTA (based on baseline characteristic) against dummies for leniency quintile and form an F -statistic for the test of the null that these dummies all equal 0; these are compared to a distribution of F -statistics produced by permuting the leniency quintile dummies randomly within each cell. The figure graphs the resulting p -value distribution. See [Online Appendix A](#) for more details.

Policy evaluation – leniency design

- Check if model trained only on most lenient (least selection) can predict for less lenient who might condition on Z
- No evidence of changes in prediction quality

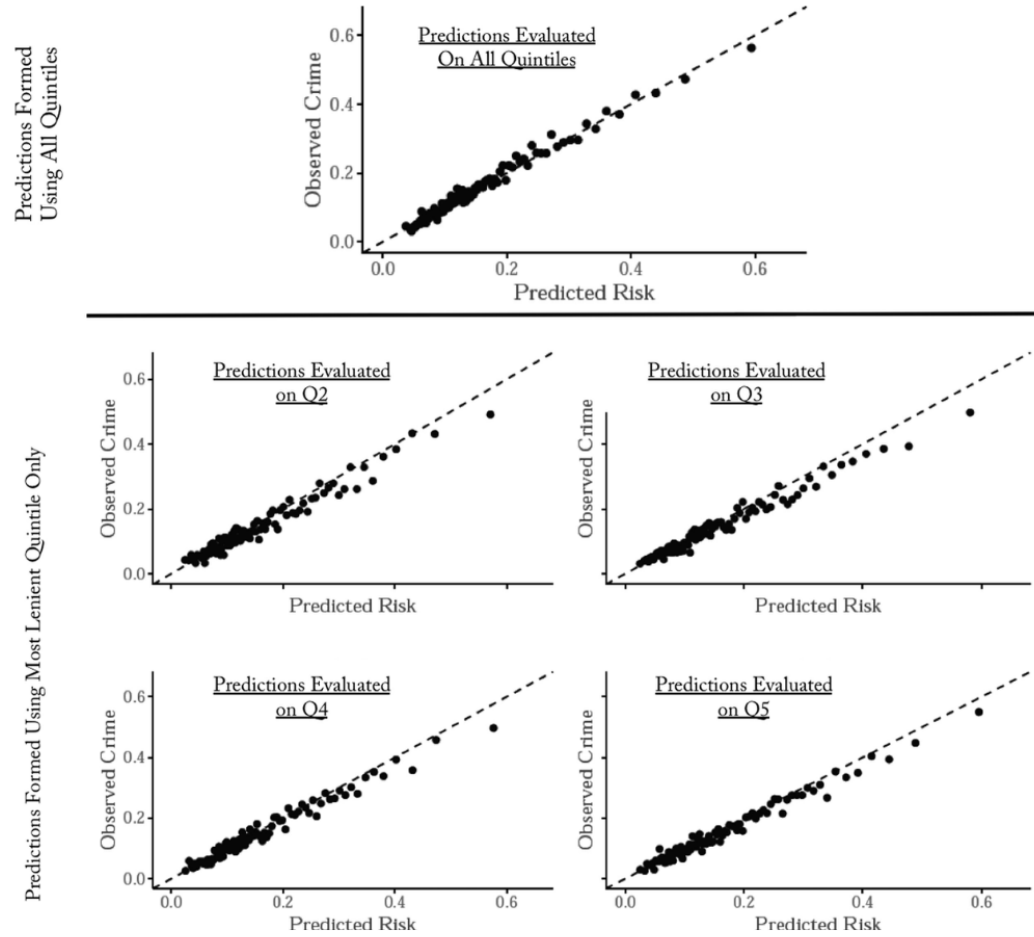


FIGURE IV

Do Judges of Different Leniency Screen Differently on Unobservables? Evaluated Predictors Formed Using Most Lenient Quintile on Other Quintiles

Policy evaluation – leniency design

- Compare effect of switch leniency groups to algorithm counterfactuals
 - Lower jail rate to get same crime rate
 - Lower crime rate to same jail rate

TABLE III
DOES JAILING ADDITIONAL DEFENDANTS BY PREDICTED RISK IMPROVE ON JUDGES?
CONTRACTION OF THE MOST LENIENT JUDGES' RELEASED SET

	Judges Relative to most lenient quintile		Algorithm To achieve judge's	
	Δ Jail	Δ Crime	Δ Crime Δ Jail	Δ Jail Δ Crime
Second quintile	0.066	−0.099	0.028	−0.201
Third quintile	0.096	−0.137	0.042	−0.269
Fourth quintile	0.135	−0.206	0.068	−0.349
Fifth quintile	0.223	−0.307	0.112	−0.498

Notes. This table reports the results of contrasting the cases detained by the second through fifth most lenient quintile judges compared with the most lenient quintile judges, and to a release rule that detains additional defendants in descending order of predicted risk from an algorithm trained on failure to appear. The first column shows from where in the predicted risk distribution each less lenient quintile's judges could have drawn their marginal detainees to get from the most lenient quintile's release rate down to their own release rate if judges were detaining in descending order of risk. The second column shows what share of their marginal detainees actually come from that part of the risk distribution. The third column shows the increase in the jail rate that would be required to reach each quintile's reduction in crime rate if we jailed in descending order of the algorithm's predicted risk, while the final column shows the reduction in crime that could be achieved if we increased the jail rate by as much as the judge quintile shown in that row.

Policy evaluation – reranking and imputation

- Assume we can impute the missing outcomes for retained defendants
 - $E[y|x, \rho_j = 1]$ if above lenient judge threshold
 - $\min\{1, \alpha E[y|x, \rho_j = 1]\}$ otherwise

EFFECT OF RELEASE RULE THAT RERANKS ALL DEFENDANTS BY PREDICTED RISK
BOUNDS UNDER DIFFERENT ASSUMPTIONS ABOUT UNOBSERVABLES

Assume $y = \min(1, \alpha \hat{y})$ for additional releases beyond most lenient judge quintile's release rate							
	Value of α						
	1	1.25	1.5	2	3	...	∞
Algorithm's crime rate	0.0854	0.0863	0.0872	0.0890	0.0926		0.1049
at judge's jail rate	(0.0008)	(0.0008)	(0.0008)	(0.0009)	(0.0009)		(0.0009)
Percentage reduction	-24.68%	-24.06%	-23.01%	-21.23%	-18.35%		-14.39%
Algorithm's jail rate	0.1531	0.1590	0.1642	0.1733	0.1920		0.2343
at judge's release rate	(0.0011)	(0.0011)	(0.0011)	(0.0011)	(0.0012)		(0.0013)
Percentage reduction	-41.85%	-40.13%	-38.37%	-34.87%	-29.36%		-18.51%

Admission as prediction policy

Research in algorithmic policy

- State-of-the-art
 - Comparison of machine vs. human
 - E.g., Kleinberg et al. (2018, QJE) or Rambachan (2024, QJE)
 - Compare human decisions with counterfactual machine decisions
- Open questions
 - Are (standard) prediction algorithms powerful enough for complex data?
 1. Pre-aggregate data into tabular format – apply standard machine learning
 2. Improved policy using ChatGPT or similar model?
 - Could such predictive systems be integrated in allocative mechanisms, e.g., for centralized admission?

An elusive target

Answering the open questions require:

- An outcome of high policy relevance:
 - Outcome can be controlled by policymaker
 - State of outcome matters - monetary or non-monetary consequences
- Granular predictions of humans and/or current policies
- Hard prediction problem of sufficient scale
 - Standard algorithms cannot handle sequential, high dimensional input
 - Large scale data set for training algorithm

Introduction

We ask

- Can machine learning *outcompete current human-based policies and improve fairness?*
- Can new ML models (transformer in particular) *outperform standard models with non-sequential models* in prediction policy scenarios?
- What about the ethics of data processing – is more data better?

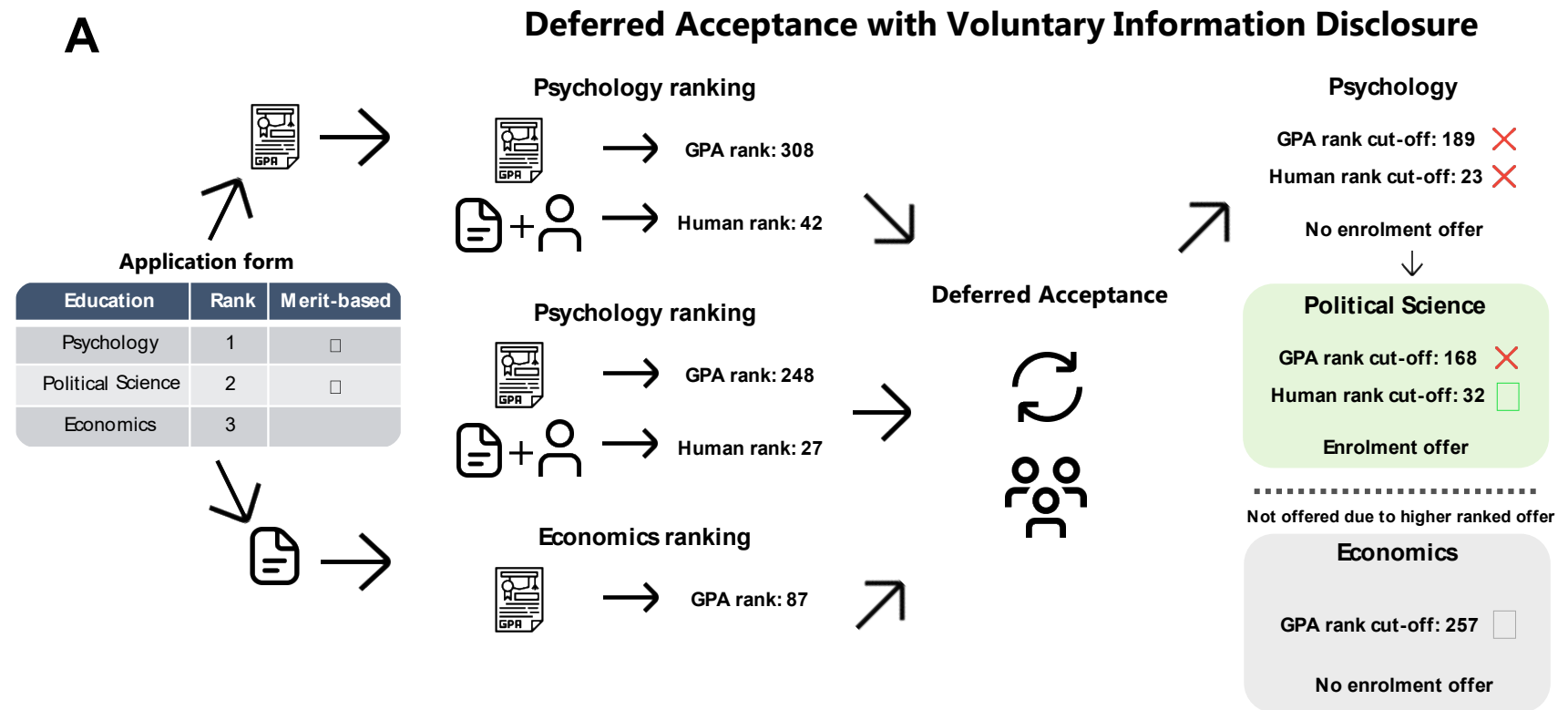
Danish higher educ. admission: policy relevance & complexity

- Centralized admission – admission criteria determines ranking
 - Primary quota (~85%): admission by high-school GPA as criteria
 - Secondary quota (~15%): faculty assessment as criteria
 - Rankings in quotas can be replaced with machine ranking
- High stakes – students waste time, programs money (approx. \$50,000 per student completed)
- Hard prediction task: no signal in grade transcripts beyond point average
- Registry data – historical data of high quality and large scale

Danish higher educ. admission: allocation system

- Admissions work using Deferred Acceptance

- Implication: admit those with highest *eligibility*
- Can view *eligibility* as predicted “risk score”



The methodological problem at hand

Data covers all applicants to higher education in the period 2006-2017

- Grades for primary and secondary school
- Applications to higher education

Repeated observations of the same student

- Hard to encode

Transformers

- Becoming a steady part of machine learning methods and everyday life
- Originally architecture is made for sentence inputs

Input and sequences of events

Input issues

- Unknown amount of grades
- Differences in courses

Similar to sentences

- Unknown sentence length
- Heterogeneity in word usage

B

Socio-demography			
Gender	...	Immigrant	Birth year
Female	...	Yes	1998

Grade transcript				
Year	School	Type	Course	Grade
2011	ID1	Primary school	Math	C
⋮	⋮	⋮	⋮	⋮
2016	ID2	High school	English	A

Enrolment			
Year	Education	Rank	Status
2017	Psychology	1	Applied
2017	Political Science	2	Enrolled
2017	Economics	3	Applied

Sequence type	Relative year	>	>	>	0	>	...	>	5	>	6	>	6	>	6
	Age	>	>	>	13	>	...	>	18	>	19	>	19	>	19
	Demography	>	>	>	Female	>	...	>	Immigrant	>		>		>	
	School	>	>	>		>		>		>		>		>	
	Type	>	>	>	ID1	>	...	>	ID2	>		>		>	
	Course	>	>	>	Primary school	>	...	>	High school	>		>		>	
	Grade	>	>	>	Math	>	...	>	English	>		>		>	
	Education	>	>	>	C	>	...	>	A	>		>		>	
	Rank	>	>	>		>		>		>		>		>	
	Status	>	>	>		>		>		>		>		>	

Two types of models

We create three different input types

- Baseline: only GPA and program
- Academic: enrolment place and transcripts
- All: Full application and background information

Why create different models?

- Separate gains from models and information sets (input)
 - Background information may encode bias, no legal basis for using

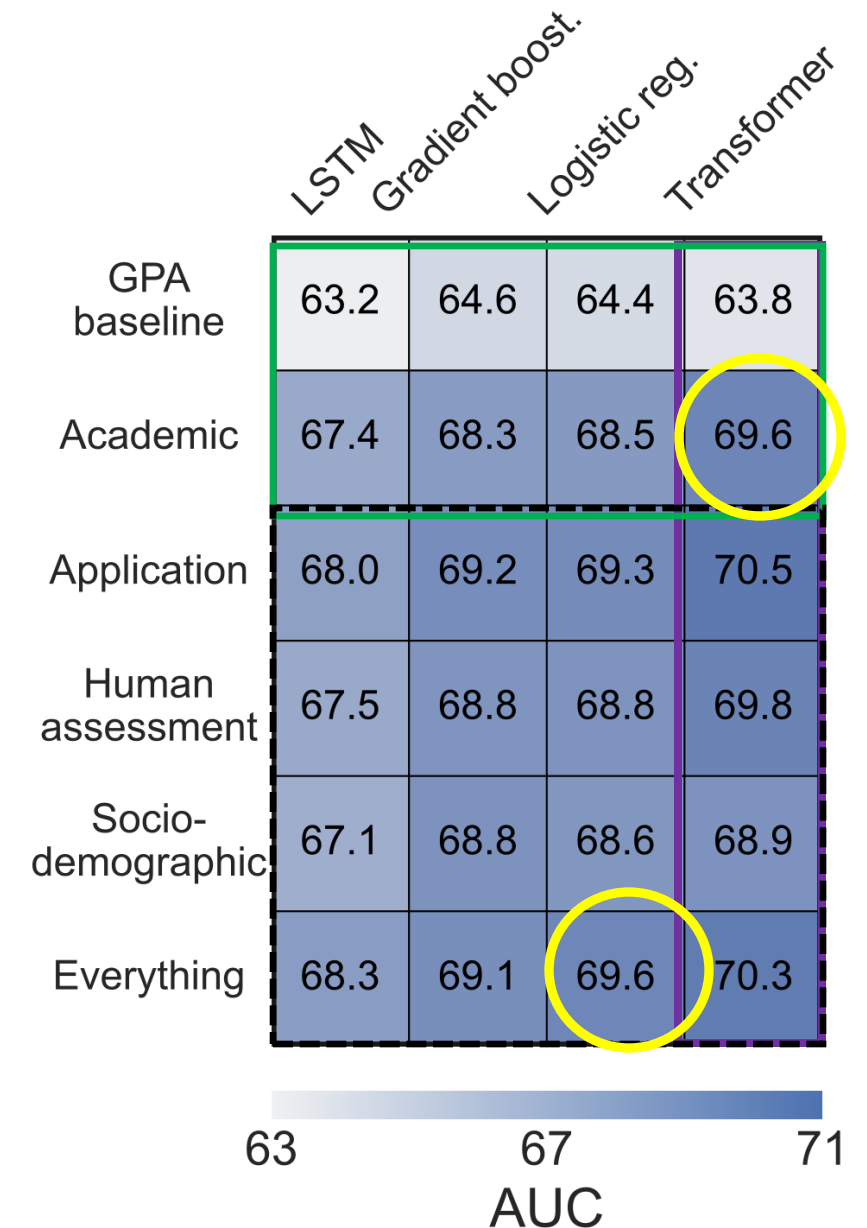
Performance across program

We use Area under the ROC Curve (AUC) to measure performance

- Higher is better (50% is random guess, 100% is perfect accuracy)
- Probability that the model will rank a randomly chosen positive instance higher than a randomly chosen negative instance

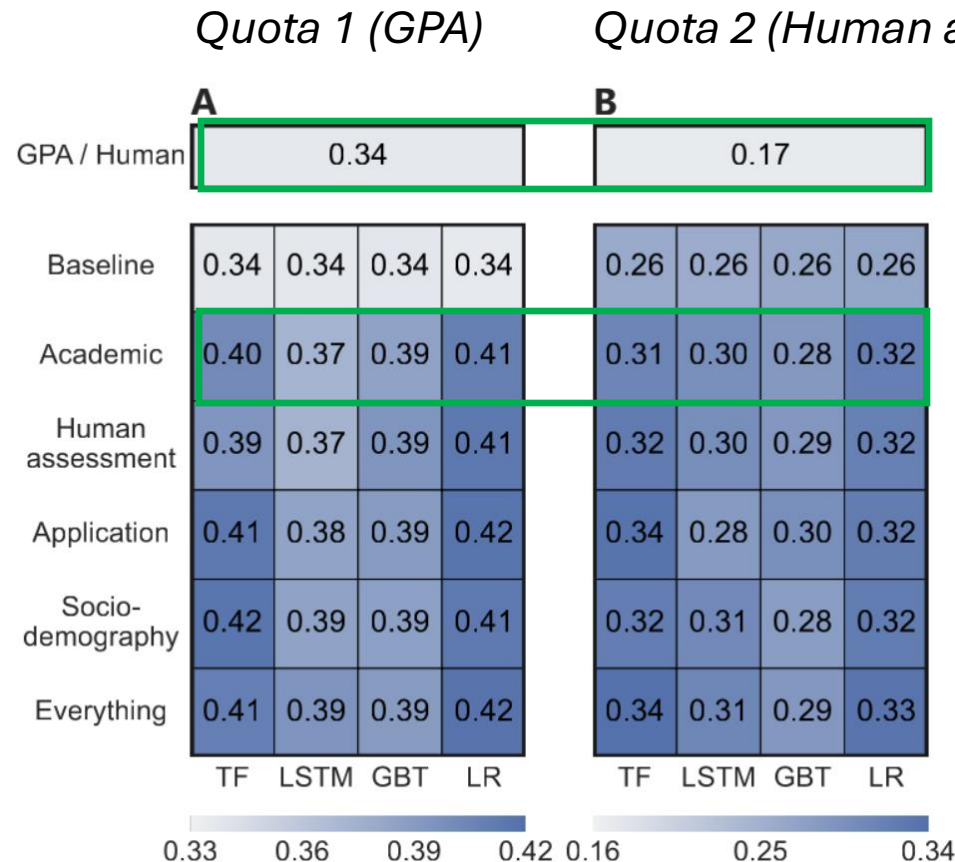
Insights

- ML outperforms current admission (quota 1)
- Transformer outperforms traditional ML:
Gain equal to gain from protected attr.



Performance on other outcomes

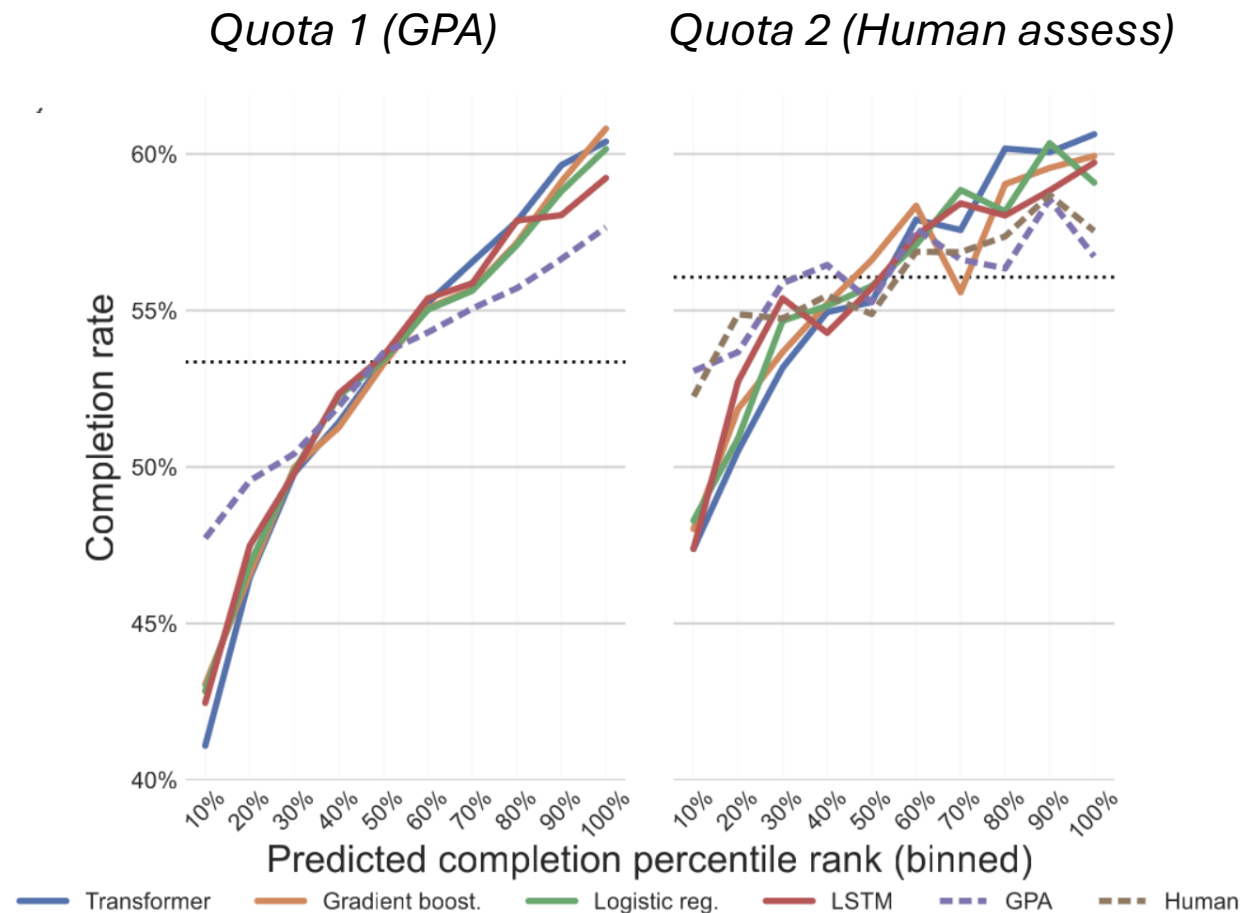
Are predictions correlated with another outcome, e.g., grades obtained?



Performance within program

Rank all students based on predicted completion

- Ranking is done within each institution (split by by decile)



Hypothetical admission scenario

New DK policy from 2023: Cut 10% of intake going forward

Can evaluate how to target students using admission entry

- Ignore substitution effects – consider only actual program
- Within program metrics are sufficient

Takeaways:

Better at identifying at-risk students than

- Current systems, both grade-based (GPA) and faculty screening
- Traditional machine learning methods

Algorithmic fairness

Trust in public policy depends on impartial treatment.

- Goal: Measure bias and discrimination in policies – how?
- Two common goals
 - Sufficient information: given a prediction (such as dropout risk), the likelihood of that prediction being correct should be equal for all groups
 - Equal opportunities: equal rate of admission chances across groups
- Problem – inherent tradeoff in these two measures (Kleinberg et al., 2017)
 - Alternative measure – difference in ROC-curve

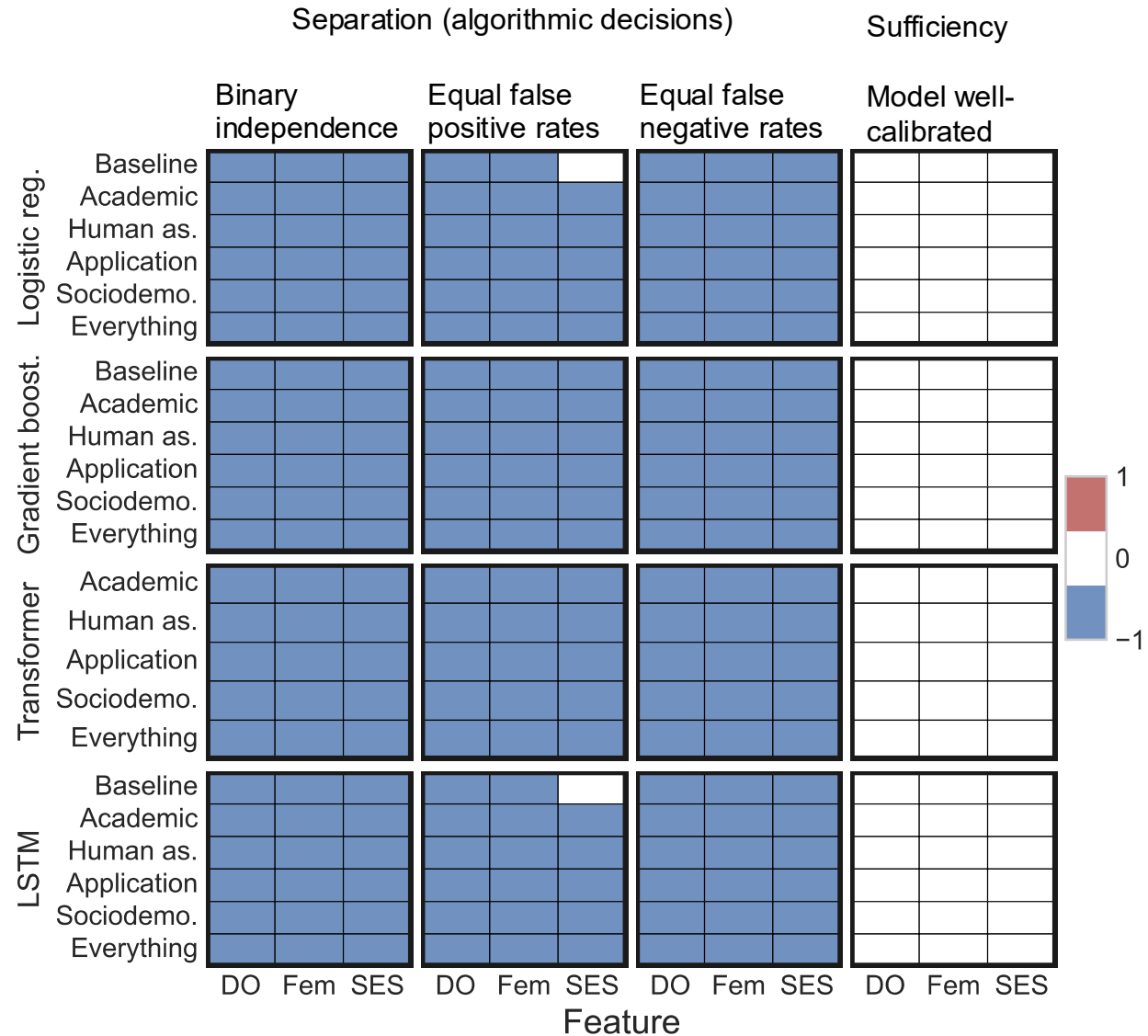
Algorithmic fairness

If the sensitive variables have predictive powers, the models have encoded it

- Across all models
- Across a wide variety of sensitive attributes

Findings

- Groups are not treated equally (*separation*)
- No fairness through unawareness (*sufficiency*)



Fairness compared to current systems

Compare models and current systems

- Use that each institution has a ranking of students used for admission

Fairness computed using ABROCA

- ABROCA measures difference in performance
 - ‘How good are the systems at ranking applicants’ across sensitive attributes
- Lower values indicate a more fair model

Transformer and LSTM better than Q1

A				B				
LR	GPA / Human	13.7	10.9	9.5	16.5	13.4	13.7	
	Baseline	13.3	10.8	9.2	14.4	14.4	13.6	
	Academic	12.8	10.3	9.7	14.2	13.6	14.4	
	Everything	13.2	11.1	10.3	14.9	13.7	14.3	
GBT	Baseline	13.0	10.7	9.6	17.0	14.3	13.5	
	Academic	13.2	10.2	9.8	17.1	12.3	13.9	
	Everything	12.9	10.8	10.1	14.7	12.5	12.6	
TF	Baseline	13.1	10.8	9.4	20.4	12.9	13.2	
	Academic	11.9	9.6	9.7	19.3	13.4	13.5	
	Everything	12.5	10.5	9.9	18.3	13.8	14.2	
LSTM	Baseline	10.7	8.8	7.7	14.9	10.0	11.3	
	Academic	11.1	8.7	8.4	14.6	10.4	10.9	
	Everything	12.0	9.9	8.9	15.6	11.4	10.2	
		Fem	Orig	SES	Fem	Orig	SES	
		8	10	12	14	10	15	20
		GPA asmt. intake ABROCA			Human asmt. intake ABROCA			

Summary

Prediction can be integral in policy-making

Evaluation requires careful attention to ***selective labels problem***

- AI performance > human performance
- Problem: AI decisions support $\neq$ better decisions

Next time

- More on human AI collaboration
- Applications of chatbots
- Delegation vs. not